

Precedence-sensitive locality in English locative inversion

Zachary Feldcamp

feldcamp@mit.edu

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Abstract

This paper argues that syntactic locality is sensitive to linear order. English locative inversion shows that a PP cannot raise across a preceding DP pivot, even when that DP is structurally lower than the PP; PP raising becomes possible only when the DP is postposed. I analyze this pattern with Left-bracket Closeness: among goals in a probe’s search domain, the closest goal is the one whose left bracket occurs first in the linearized representation. The analysis is shown to correctly predict a previously unobserved restriction on subextraction in locative inversion.

Keywords: syntax; locality; locative inversion; linear order; subextraction; intervention

1. Introduction

Research on syntactic movement recognizes some version of the minimality principle in (1), which requires movement to be as short as possible (e.g., Chomsky, 1995; Rizzi, 1990).¹ Thus, *wh*-questions in English-like languages involve a C^0 head that attracts the closest *wh*P, so that, given a [+*wh*] subject and object, the subject must move to

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spec,CP (2a), intervening for movement of the object (2b) (Chomsky, 1973). Syntactic intervention effects have also been described for plausible instances of A-movement, e.g., in superraising (Rizzi, 1990), in object shift (Vikner, 1991), and in specificational copular sentences (den Dikken, 1994).

(1) *Attract Closest*

Given two potential goals G and K for an attracting head P, move whichever of G or K is *closer* to P.

(2) (a) Who₁ do you think ___₁ will perform what?

(b) *What₂ do you think who will perform ___₂?

The concept of “closeness” is standardly formulated within the generative literature in purely structural terms: given G and K c-commanded by P, G is *closer* than K to P if G asymmetrically c-commands K (e.g., Rizzi, 2013). But c-command typically coincides with precedence (i.e., linear order) in intervention configurations. One might then suppose that what makes movement of the subject minimal in (2) is not that the subject asymmetrically c-commands the object, but that the subject *precedes* the object. This paper uses English locative inversion to argue that syntactic minimality is indeed computed over an ordered representation of a probe’s search domain; concretely, I argue for a locality metric that I call *Left-bracket Closeness* (3).

(3) *Left-bracket Closeness* (preliminary version)

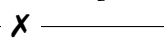
G is closer to P than K iff the left bracket of G precedes the left bracket of K in the bracketed representation of P’s search domain.

Recently, Hewett (2025) has argued that precedence plays a role in syntactic locality on the basis of configurations in which a preceding G intervenes for the movement of some K, even though G and K are not in a c-command relation. I show that his data

are compatible both with *weak* and *strong* roles for precedence. In a weak formulation, precedence is consulted only as a tie-breaker when candidates are not in a c-command relation. In a strong formulation like that in (3), precedence is not merely a tie-breaker but the primary metric of locality.

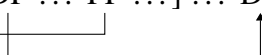
The empirical focus of this paper is a set of configurations in which a DP and a PP compete to satisfy the EPP. I show that it is ungrammatical to raise a PP across a DP that precedes it (4a). Crucially, this does not depend on their c-command relations: even when the DP is structurally lower than the PP, the DP may raise, but the PP may not (4aii). However, if the DP is postposed to the right of the PP, then it is grammatical for the PP to raise (4b).² Much of the literature on locative inversion presupposes some version of Attract Closest and attempts to explain how raising of a structurally low PP across a higher DP is possible, sometimes by relating it to the postposing of the DP (e.g., Doggett, 2004; Rizzi and Shlonsky, 2006; Storment, 2025). The missing fact is that the DP must be postposed even when it is structurally *lower* than the PP. This paradigm is explained if precedence does not merely play a weak, tie-breaking role, but is in fact central to syntactic locality, as implemented in (3).

(4) (a) *PP may not raise past preceding DP*

(i) [TP [T ... [VP ... DP ... [... PP ...] ...]] (DP underlyingly c-commands PP) 

(ii) [TP [T ... [VP ... [... DP ...] ... PP ...]] (PP underlyingly c-commands DP) 

(b) *PP raises if DP is postposed*

[TP [T ... [VP ... ~~DP~~ ... PP ...] ... DP]] 

²Postposing is represented here as rightward movement, but this is not essential to the analysis; see Section 5.

We begin in Section 2 with a locality puzzle: in some cases of locative inversion, a DP coargument of a PP blocks the PP from raising; in other cases, however, a PP may raise past a DP, even when the DP underlyingly c-commands it. In Section 3 I show that the grammatical cases involve postposing of the DP pivot, using secondary predicates as rigid diagnostic “landmarks”. Because the pivot must be postposed even when the PP is underlyingly higher, postposing cannot merely be a strategy for removing a c-commanding intervener; I argue that it is instead a strategy for removing a *preceding* intervener. In Sections 4 and 5 I introduce Left-bracket Closeness and show that it derives this generalization, unlike other locality metrics. Next, I show in Section 6 how the analysis extends to cases of locality involving one goal dominating another; in particular, I present novel data in which subextraction of a PP from a DP is blocked in locative inversion. I conclude in Section 7 with speculations on the role of postposing in hierarchy-inverting movement more generally.

2. The (non)-existence of syntactic intervention effects in PP raising

The focus of this paper is the syntactic locality of English locative inversion (LocInv). The alternation between inverted and non-inverted clauses is puzzling because of its apparent optionality: for a given assignment of theta roles to a DP and a PP, either the DP or the PP may check the EPP. To illustrate, whereas an uninverted passive dative construction like (5a) maps the DP Theme to the structural subject position and the PP Recipient to a position within the verb phrase, LocInv appears to map the PP to the subject position (5b).³ That this is the case is evidenced by a variety of subject properties, several of which are discussed by Bresnan (1994). For instance, the fronted PP

³The term “locative inversion” is intended to encompass all cases in which a broadly spatial PP moves to what appears to be the ordinary subject position, including PPs whose semantics are locative, directional, comitative, and so on. For discussion of what delimits this class of PPs, see Birner and Ward (1998, 238ff.). For specific discussion about cases like that in (5) involving PP recipients, see den Dikken (1995, 119ff.).

exhibits complementizer trace effects (6a; Bresnan, 1977), unlike, say, topicalization of a PP from a presentational *there* clause (6b).⁴ The PP may also apparently undergo subject-to-subject raising (7; Postal, 1977, p. 148); observe, moreover, that this requires the embedded clause to be non-finite (8)—the typical expectation for A-movement in English.⁵ As for the DP controller of subject agreement in (5b), which I call the *pivot*, we will see evidence in Section 3 that it occupies a postposed VP-external position.

(5) (a) *Non-inverted mapping of theta roles to structural positions*

[_{DP} Various medical supplies]_{Theme} were handed [_{PP} to the student doctor]_{Recipient}.

(b) *Inverted mapping of theta roles to structural positions*

[_{PP} To the student doctor]_{Recipient} were handed [_{DP} various medical supplies]_{Theme}.

(6) (a) To the student doctor, it seems (*that) were handed various medical supplies.

(b) To the student doctor, it seems (that) there were handed various medical supplies.

(7) To the student doctor seem to have been handed various medical supplies.

(8) *To the student doctor seem have been handed various medical supplies.

That the position of the PP in (5b) is derived via movement from within the VP is indicated by the fact that the verb *hand* subcategorizes for an obligatory PP argument (9). The optionality in EPP checking then presents a locality puzzle. If the PP is underlyingly higher than the DP, how does the PP not intervene for raising of the DP in non-inverted passive sentences, as in (10a)? And if the DP is underlyingly higher, how does it not

⁴Unless otherwise indicated, judgments reported in this paper are the author's and have been checked with several other native speakers of English.

⁵The *that*-trace effect contrast in (6) is intended to motivate the hypothesis that the PP of LocInv occupies the structural subject position. It should be noted, however, that this argument has been found unconvincing by Bruening (2010), who instead argues that the PP occupies an $\bar{A}$ -topic position. The only assumption that is essential for this paper is that the PP of LocInv is sensitive to A-movement locality conditions, which is established on the basis of DP intervention effects in Section 2.2 and in Section 6.

intervene for the raising of the PP to spec,TP in LocInv (10b)? Either movement to the subject position would seem to violate Attract Closest.⁶

(9) Various medical supplies were handed *(to the student doctor).

(10) (a) [TP DP [T' T ... [VP ... [... DP ...] ... PP ...]]]



(b) [TP PP [T' T ... [VP ... DP ... [... PP ...]]]]



The point of this section is to illustrate the challenges that LocInv poses for theories of syntactic locality. On the one hand, there do exist configurations in which a low PP may raise across a higher DP (Section 2.1). Yet there exist other configurations in which a DP pivot appears to *block* the raising of a PP. It is then descriptively inadequate to simply deny competition between a DP and a PP in raising. The task then becomes to investigate the conditions under which hierarchy-inverting movement may occur (Section 3).

2.1. Reversal of c-command relations without intervention effects

The configuration in (10b), in which PP raising involves a genuine “inversion” of c-command relations, is only problematic if it actually exists, as is assumed to be the case by Collins (1997), Doggett (2005), and Hoekstra and Mulder (1990), among others. This assumption has been called into question by recent work by Laparle (2019) and Liu (2025). It is then useful to investigate plausible cases of LocInv in which genuine inversion appears to occur. I present three such cases: (i) DP-PP small-clauses, (ii) DP-AP small clauses, where A takes PP as its complement, and (iii) depictive locative constructions.

⁶The PP in (10) is intended to range over complements, adjuncts, or specifiers. We assume that rightward attachment of non-complement PPs is *not* categorically banned (contra Kayne, 1994), a position defended by Abels and Neeleman (2012), Bruening (2018), Ernst (2004), and Pesetsky (1995), among others. That is, hierarchy does not uniquely fix linear order; rather, linearization is a separate procedure that can make reference to additional grammatical information such as featural specifications or prosodic weight.

First, typical cases of LocInv involve a relation of predication between the pivot and the fronted PP (Hoekstra & Mulder, 1990). Although we are not committed to the strong position that *all* cases of LocInv involve a DP-PP small clause, standard assumptions do motivate a small-clause analysis in copular sentences (e.g., *On the floor was a kitten*) and in particle preposing constructions (e.g., *In walked Bud*), for instance. To illustrate the widespread presence of small clauses in LocInv, let us consider the more complex case in (11b). We may construct a Beck and Johnson (2004)-type argument from the restitutive interpretation of *again* to establish the existence of an underlying DP-PP small clause. Imagine that a car accident has caused John to lose his spleen, which a surgeon has fortunately managed to replace. Even assuming (reasonably) that John has never had a spleen implanted previously, we can utter either (11a) or (11b), which have the paraphrase in (12). This is captured if *again* is adjoined to a small-clause where the DP is the subject and the PP is the predicate, as in (13).

- (11) (a) A functioning spleen has successfully been implanted in John's body again.
(Non-inverted passive; ✓restitutive)
- (b) In John's body has successfully been implanted a functioning spleen again.
(LocInv; ✓restitutive)
- (12) There was a successful operation of implanting a spleen in John's body, and there was a spleen in John's body before.
- (13) [_{PREDP} [_{PRED'} [_{DP} a functioning spleen] [_{PRED'} Pred [_{PP} in John's body]]] again]

The small clause is analyzed in (13) as a PredP, involving asymmetric c-command between DP and PP. Doggett (2004) argues for a high DP/low PP structure in cases like (11b) on the basis of Barss and Lasnik (1986)-type left-to-right asymmetries in active-voice configurations where the DP and the PP are apparently in situ. For instance, a downward-entailing DP may license a negative-polarity PP to its right (14a), but not vice

versa (14b), unless a downward-entailing PP raises (14c). If the small clause underlying (11b) could, hypothetically, involve the PP c-commanding the DP, then we would expect that the PP could take scope over the DP; examples such as (14) suggest that this is not the case.

- (14) (a) The surgeon implanted [_{DP} no organ] [_{PP} in any part of John's body].
 (b) *The surgeon implanted [_{DP} any organ] [_{PP} nowhere in John's body].
 (c) [_{PP} Nowhere in John's body] was implanted [_{DP} any organ].

That being said, DP-XP small clauses have traditionally been analyzed as projections of the XP predicate, with the DP merged as a specifier of that projection (e.g., Matushansky, 2019; Stowell, 1983). On such an analysis, the DP and the predicate do not stand in an asymmetric c-command relation. As a second case that establishes DP-over-PP c-command independently of the internal architecture of the small clause, we may choose the predicate to be an AP whose A⁰ head selects the PP. For example, *available* l-selects for a PP complement that is headed by *to* (15a). I find that, if the PP is independently able to undergo LocInv—as a *to*-phrase is; recall example (5)—then it may also undergo LocInv when base-generated within an AP (15b). However, when *available* is replaced by an adjective that l-selects for a different P, LocInv of a *to* PP is deemed ungrammatical (16). It is therefore plausible that both the uninverted and inverted cases involve the base structure in (17).⁷ Example (15b) is then an instance of (10b), in which genuine inversion occurs.

- (15) (a) Various medical supplies are made available to/*of the students.
 (b) To the students are made available various medical supplies.
 (16) (a) Various woodworking supplies are made safe for/*to/*of the students.

⁷The reader may notice that the DP pivot is linearized to the right of the AP in (15b). This is due to a pivot postposing condition on LocInv to be discussed at length in Section 3.

(b) *To the students are made safe various woodworking supplies.

(17) [VP make [SC [DP various medical supplies] [AP available [PP to the students]]]]

The final case of a PP raising over a DP to be discussed here comes from *depictive* locative constructions. Typical locative adjuncts like the PP in (18a) are *event-oriented* (or “Davidsonian”), specifying the location of an event. The meaning of such PPs is preserved when the event is referred to anaphorically, as in (18b) (Davidson, 1967). In contrast, what I call *depictive* locatives specify the location of a particular event participant throughout the course of the event.⁸ To illustrate, the most accessible interpretation of the locative PP in (19a) describes the location of the DP internal argument—not of the activity of the event itself. Unlike Davidsonian locatives, event anaphora renders depictive locatives ungrammatical, so that the continuation in (19b) is infelicitous if the intent is to describe the location of the food specifically.

(18) (a) The chef served the final course *in the kitchen*. (Davidsonian locative)

(b) The chef served the final course. It happened in the kitchen.

(19) (a) The chef served the final course *in a handmade bowl*. (Depictive locative)

(b) The chef served the final course. #It happened in a handmade bowl.

One might suspect that depictive locatives are adjoined to NP, but this is not the case, because they are grammatical even when the DP is pronominal *it* (20a), which generally cannot be modified by adjuncts (20b).

(20) (a) The chef served it in a handmade bowl.

(b) *It in a handmade bowl was served.

(cf. *It was served in a handmade bowl*.)

⁸Maienenborn (2001) subsumes depictive locatives under the broader class of “internal” locatives.

Instead, Maienborn (2001) argued that depictive interpretations arise when a PP is merged “internal” to the verb phrase, before a DP argument is merged, resembling low-attachment analyses of AdjP depictives (e.g., Cormack and Smith, 1999; Pytkäinen, 2008). Observe, moreover, that depictive locative PPs may raise in LocInv (21). If a Maienborn-style analysis is correct, LocInv in (21) involves the base-generated structure in (22): the DP asymmetrically c-commands a PP, yet the PP crosses over the DP in movement out of the verb phrase, inverting the underlying c-command relations.

(21) *In this handmade bowl* will be served the final course of tonight’s meal.

(22) PP ... [_{VP} [_{DP} the final course] [_{V'} serve [_{PP} ~~in this handmade bowl~~]]]


2.2. Reversal of c-command relations with intervention effects

We have seen evidence that LocInv can result in the reversal of the underlying c-command relations between a DP and a PP, as schematically represented in (10b). There is an argument, however, that a DP *can* block the raising of a PP, due (in somewhat different forms) to Rezac (2006) and Stowell (1981), who examined the circumstances in which a pivot may occupy the position to the immediate left of a participial verb (23).⁹ What they observed was that in a sentence involving a *there* expletive subject and a PP topic, the pivot may either be immediately preverbal (23a) or occupy a postverbal position (23b); when LocInv occurs, however, only the postverbal position is available (23c-d).

(23) *Stowell-Rezac paradigm*

(a) To the general there was *a medal* awarded. (*there* + PPTop; preverbal pivot)

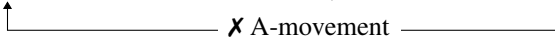
(b) To the general there was awarded *a medal*. (*there* + PPTop; postverbal pivot)

(c) *To the general was *a medal* awarded. (LocInv; preverbal pivot)

⁹To my knowledge, contrasts like that in (23) were first noted by Milsark (1974) in his discussion of the proposal of Kuno (1971) that locative inversion and *there*-existentials are derivationally related.

- (d) To the general was awarded *a medal*. (LocInv; postverbal pivot)

Rezac argued that the contrasts in (23) can be explained under an analysis of LocInv in which the PP undergoes A-movement to subject position akin to the raising of a DP: (23c) is ungrammatical because the pre-participial DP occupies spec,vP, asymmetrically c-commanding the PP in the VP, so that raising the PP to spec,TP violates Attract Closest, as illustrated in (24); in contrast, the PP movement in (23a) is $\bar{A}$ -movement, which is assumed not to be subject to intervention by DPs (25). There are alternative analyses of LocInv due to Postal (2004) and others that involve $\bar{A}$ -movement of the PP to a topic position where it can license a null expletive counterpart of *there* in subject position; but these alternatives incorrectly predict the null expletive version of (25) to be grammatical, for (23a) and (23c) would be analyzed as being structurally identical. Null expletive analyses are discussed further in Section 6.

- (24) [TP PP₁ [T' was [vP [*a medal*] [v' v [VP awarded *t*₁]]]]]


- (25) [CP PP₁ [C' C [TP there [T' was [vP [*a medal*] [v' v [VP awarded *t*₁]]]]]]]


We are faced with a puzzle: the Stowell-Rezac paradigm presents a case in which a c-commanding DP appears to block the raising of a PP, yet other configurations involve the raising of a PP over a DP without intervention effects. Next, we will see evidence that the linear position of the DP and the PP is critical in enabling the PP to raise.

3. Precedence-sensitive intervention effects in PP raising

We would like to understand why a DP in a preverbal position intervenes for the raising of a PP (23c), while a postverbal DP does not necessarily do so (23d). Towards

this goal, I argue that a DP intervenes when it *precedes* the PP; this is stated as a descriptive generalization in (26). In particular, we will see that a preceding DP blocks a PP from raising even when the DP is structurally *lower* than the PP. I argue, then, that pivot postposing should not be understood as a mechanism to obviate c-command Attract Closest; for analyses based on such an assumption undergenerate by incorrectly predicting a structurally low DP pivot to be able to remain in situ when a high PP raises.

(26) *Pivot postposing condition on English locative inversion*

A PP cannot raise over a preceding DP. In order for a PP to raise, the DP pivot must be linearized to the right of the verb phrase containing the PP.

3.1. Background on pivot postposing

Let us begin by examining the position of pivots that do *not* intervene for PP raising; we consider two hypotheses. Under the simplest analysis, the postverbal pivot in cases of non-intervention like (23d) occupies its theta position. Rezac (2006) argues that this yields a structure like that in (27a) in which the DP internal argument does not c-command the recipient PP, so that the DP is not expected to intervene for raising of the PP. (How the PP does not intervene for raising of the DP in the ordinary passive is not discussed.) Doggett (2004) and Stowell (1981), among others, however, do not assume that the PP underlyingly c-commands the pivot in all relevant configurations; raising of the PP over the DP then poses a locality problem. Such work proposes various mechanisms by which movement of the pivot to a right-peripheral position enables the PP to raise (28). Both analyses correctly allow the pivot to be pronounced to the right of the ordinary subject position.

(27) *In-situ pivot analysis*

(Culicover and Levine, 2001; Rezac, 2006)

[_{VP} [_{V'} award [_{DP} a medal]] [_{PP} to the general]]

(28) *Postposed pivot analysis* (Doggett, 2004; Rochemont, 1978; Stowell, 1981)

[_{VP} [_{V'} v [_{VP} t₁ [_{V'} award [_{PP} to the general]]]] [_{DP} a medal]_i]

Let us consider existing arguments for postposing. Iwakura (1978) observed that a DP subject ordinarily precedes auxiliaries and preverbal adverbs (29a), whereas it must follow the lexical verb in LocInv (29b,c). Iwakura argued on the basis of such contrasts that LocInv cannot simply be derived by a combination of PP fronting and head-movement of a verb, with the subject remaining in the EPP position throughout the derivation, as Langendoen (1973) proposed for simple past instances of LocInv like (30). Instead, Iwakura argued that LocInv involves both PP fronting and an operation of *Subject Postposing*, the latter operation lowering the subject from its spec,TP to a right-hand VP adjunct position.

(29) (a) *A large poster of Lincoln* has long stood in each hallway.

(b) *In each hallway has *a large poster of Lincoln* long stood.

(c) In each hallway has long stood *a large poster of Lincoln*. (Iwakura, 1978, 334, fn. 11)

(30) In the Italian garden stands *an elegant fountain*. (Langendoen, 1973, 30, ex. 16)

A subject lowering operation was attractive in a theory of clause-structure where the subject is base-generated in the EPP position, as was assumed by Iwakura (1978), Rochemont (1978), and Stowell (1981), and in other early analyses of LocInv. Once one adopts the VP-internal subject hypothesis, however, it is no longer obvious that such an operation is needed. It has been suggested, for instance, that the DP pivot may simply remain in its VP-internal theta position (Culicover & Levine, 2001; Rezac, 2006), perhaps in conjunction with head-movement of V to a low position in the aspectual domain that precedes the theta positions of arguments (Salzmann, 2013). That is to say, the fact

that the pivot is pronounced to the right of the verb in LocInv does not itself constitute evidence for postposing.

But the hypothesis that LocInv involves right-adjunction of the pivot has long received independent motivation from the observation that the pivot shares similar prosodic and information-structural properties with heavy NP shift (HNPS), as argued by Rochemont (1978, 1986). The claim of such work is that the pivot in (29c) occupies the same “constructional focus position” as in instances of HNPS like (31). To see that the pivot is focused, notice that in response to a *wh*-question, it felicitously corresponds to the *wh*-phrase of the question (32a), but does not felicitously correspond to unfocused material (32b). Note also that, as in HNPS, the acceptability of LocInv improves as the DP becomes prosodically heavier (33).¹⁰

- (31) The superintendent placed —₁ [in each hallway] [a large poster of Lincoln]₁.
(HNPS)

- (32) (a) Q: What artworks are in each hallway?

A: In each hallway have long stood these large posters of Abraham Lincoln.

- (b) Q: Where are these artworks?

A: #In each hallway have long stood these large posters of Abraham Lincoln.

- (33) (a) In each hallway were placed *them/?these posters/these large posters of Abraham Lincoln.

- (b) The superintendent placed —₁ [in each hallway] [*them/?these posters/these large posters of Abraham Lincoln]₁.

What is of interest here, however, is not merely whether the pivot of LocInv is focused, but whether it is in a position to the right of where it is base-generated, in which

¹⁰See Bresnan (1994) for a review of the “presentational” focus properties of the pivot in LocInv.

case it would resemble HNPS even more closely. In support of this pivot postposing hypothesis, Doggett (2004) and Pesetsky (1994) show that LocInv requires the pivot to occur to the right of verb particles (34) and that there is a somewhat weaker preference for the pivot to occur after PP recipients (35). However, data from particle positions are also compatible with an analysis in which the pivot remains in situ and the particle must be reanalyzed with the verb when the PP raises, as den Dikken and Næss (1993) suggest in connection with examples like (34). Moreover, various authors have observed that the pivot may precede right-adjoined adverbs (36) and PPs (37), suggesting that the pivot may remain in situ in LocInv. Notably, Culicover and Levine (2001) have argued that “heavy” inversion involves postposing of the pivot, whereas “light” inversion allows the pivot to remain in situ in cases like (36).

(34) To the stockholders were sent ⟨*several brochures⟩ out ⟨several brochures⟩. (Pesetsky, 1994, ex. 19)

(35) In this room were assigned ⟨??several important cases⟩ to him ⟨several important cases⟩. (ibid. ex. 20)

(36) Into the room walked Robin carefully. (Culicover & Levine, 2001, 292 ex. 18b)

(37) In this hall will be awarded one prize to a linguistics student. (Rezac, 2006, 692, ex. 17a)

I suggest that these apparently conflicting diagnostics from word order are not as useful as one might like for determining the position of the pivot, because run-of-the-mill PPs and AdvPs are relatively free in their right-adjoined positions. An AdvP adjunct like *reluctantly* in (38), for instance, may be ordered more or less freely relative to other postverbal adjuncts like *this morning*. I assume that this is because such modifiers may attach to various right-hand VP/vP adjunct positions (e.g., Ernst, 2004).

(38) A. Did Amelia take her pills?

B. Yes, she took them ⟨reluctantly⟩ this morning ⟨reluctantly⟩.

Examples like (36) and (37) then do not convincingly distinguish between an in-situ pivot analysis and a postposed pivot analysis, since the adjunct could be attaching to the right of the derived position of the pivot. In order to diagnose the position of the pivot, then, I instead consider its position relative to phrases that are more or less fixed in their VP-internal positions.

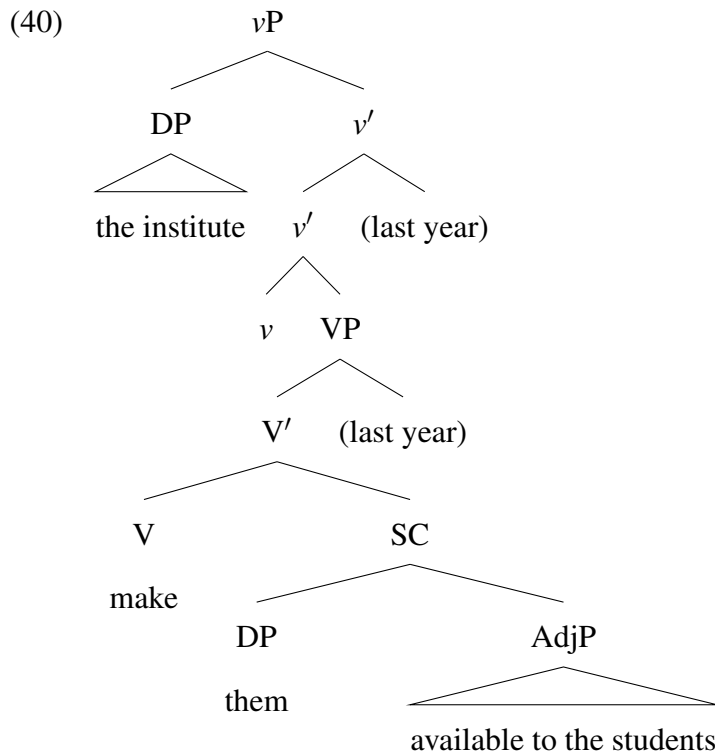
3.2. Diagnosing pivot postposing with secondary predicates

In order to diagnose the position of the pivot more accurately than earlier word-order diagnostics, we must use a “landmark” that is fixed low in the VP (to use a term of Andrew Carnie’s). If the pivot may be pronounced to the left of that landmark in LocInv, then it likely remains in situ, or at least does not participate in a detectable postposing process. Alternatively, if the pivot may ordinarily appear to the left of that landmark, say, in transitive verb phrases, but must appear to the right of it in LocInv, then postposing of the pivot must be a necessary part of the derivation of LocInv.

I suggest that AdjP predicates of VP-internal small clauses are landmarks of the desired kind, as diagnosed by the fact that they cannot follow right-hand adverbials. Thus, we see in (39) that the secondary predicate *available to the students* cannot follow the temporal adjunct *last year*. This is captured by an analysis in which the secondary predicate occupies a fixed, structurally low position in a small clause complement of V, whereas adverbials adjoin to projections of the verb, as in (40).

(39) A. What did the institute end up doing with the remaining medical supplies?

B. The institute made them ⟨available to the students⟩ last year ⟨*available to the students⟩.



We can then use AdjP secondary predicates as landmarks for diagnosing the position of the pivot in LocInv. Under the in-situ pivot analysis, we predict that the pivot should ordinarily precede an AdjP secondary predicate (although it may follow the AdjP when the heaviness and focus conditions of HNPS are met), as in transitive sentences such as (41a). What we see in (41b-c) is the opposite. LocInv is unacceptable when the pivot occurs to the left of the small clause predicate *available*, but improves when the pivot is prosodically heavy and pronounced on the *right*. Such sentences are most naturally pronounced with a prosodic break after the secondary predicate and with prominent accents on both the predicate and the postposed pivot, as approximately represented in (42). A parallel contrast is observed in (43).

- (41) (a) The institute has made *the following medical supplies* available to the students.
 (b) *To the students have been made *the following medical supplies* available.
 (c) To the students have been made available *the following medical supplies*.
- (42) (To the students have been made aVAIlable) || (the following MEdical supplies)
- (43) (a) The airline has left *a seat* unoccupied in the first-class cabin.
 (b) *In the first-class cabin has been left *a seat* unoccupied.
 (c) In the first-class cabin has been left unoccupied *a seat reserved for the captain's daughter*.

The paradigms in (41) and (43) bear out the predictions of the pivot postposing analysis. The diagram in (44) for the vP of the passive LocInv sentence in (43c) offers one possible implementation, involving rightward movement of the pivot to spec,vP (an alternative remnant movement derivation is considered in Section 5.4). If the pivot remained in situ as in (45), we would incorrectly predict that it could be pronounced to the left of the embedded AdjP predicate, as in ordinary transitive sentences.

- (44) [_{vP} [_{v'} be [_{VP} [_{v'} left [_{SC} *t*₁ unoccupied]] [_{PP} in the cabin]]] [_{DP} *a seat RC*]₁]
- (45) *[[_{vP} [_{v'} be [_{VP} [_{v'} left [_{SC} [_{DP} *a seat*] unoccupied]] [_{PP} in the cabin]]]]
-
- The diagram shows a horizontal line from the *t*₁ in the SC of (44) to the *a seat RC* in the spec,vP position, with an upward arrow at the destination. In (45), the *a seat* is inside the SC, and no such movement is indicated.

3.3. Postposing as a locality obviation strategy

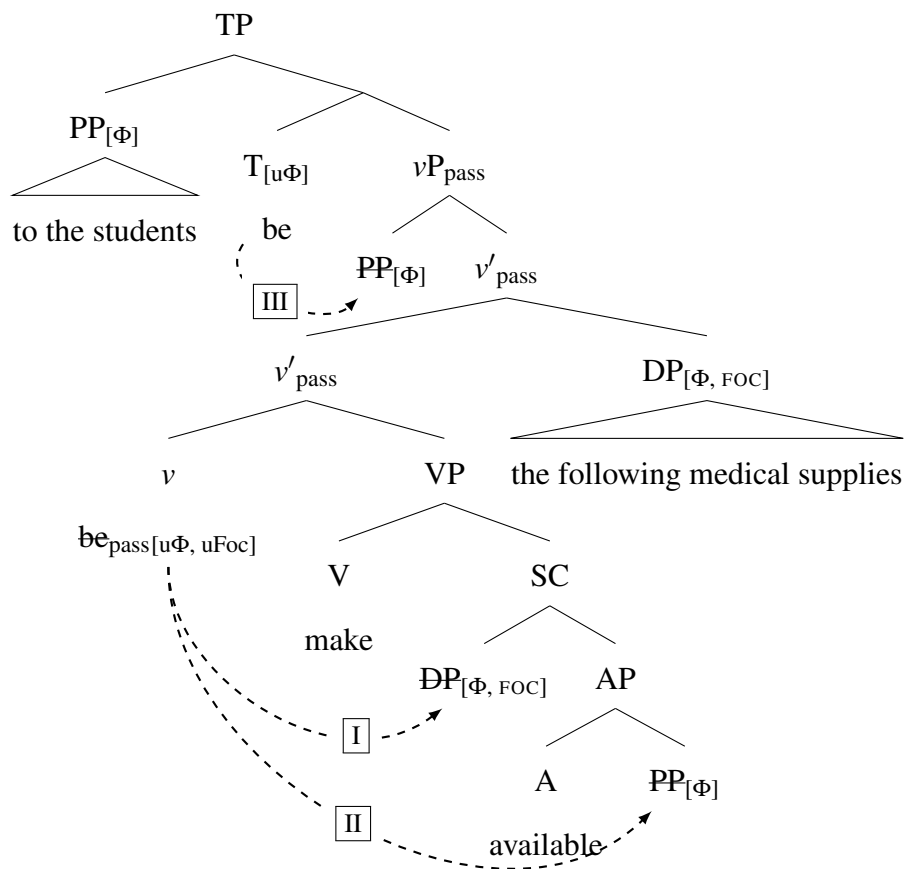
We have seen that raising a PP requires the DP pivot to meet three conditions:

- (i) the pivot must be *postposed*, or pronounced to the right of its ordinary theta position;
- (ii) the pivot must be focused; and
- (iii) there is a somewhat weaker and perhaps gradient condition that the pivot must be prosodically heavy. These are not conditions on PP movement in general, nor do they all necessarily go hand-in-hand elsewhere in syntax.

Several authors have explained these conditions as the side effects of what I call a “locality obviation” strategy, that is, a sequence of movements which enable a PP to circumvent an otherwise intervening DP and raise to the EPP position (e.g., Doggett, 2004; Rizzi and Shlonsky, 2006; Storment, 2025). These authors observe that LocInv may involve an underlying configuration in which the DP asymmetrically c-commands the PP, as we have seen in Section 2.1; all else being equal, then, Attract Closest should prevent the PP from raising to the EPP position. In order to derive PP raising in such cases, they propose that certain intermediate movements allow the PP to be at least as close to spec,TP as the DP.

The leapfrogging analysis of Doggett (2004, 2005) may be used as a representative example of how the pivot conditions may be explained as the side effects of such a strategy. Consider the derivation of (41c) in (46), where the DP pivot underlyingly c-commands the PP. Doggett assumes that v may bear two probes, [uFoc] and [u ϕ]. To derive LocInv, the [uFoc] probe first attracts a focused DP to a right-hand specifier I; the existence of this probe is independently motivated by HNPS. A second [u ϕ] attracts a [ϕ]-bearing XP (here, PP) to an *outer*, left-hand specifier of v P II; this probe is independently implicated in the derivation of preverbal pivots in passive expletive constructions of the form in (23a). This results in a v P where the PP c-commands the DP. T then identifies the PP as its goal III.

(46) *Leapfrogging strategy of obviating a c-commanding intervener*



Doggett’s analysis has the virtue that, rather than stipulating focus postposing as a construction-specific quirk of LocInv, it shows that a sequence of independently motivated movements may be used to enable a PP to raise over an intervening DP. The focus, heaviness, and rightward linearization conditions on the pivot are derived as accidental side effects of these movements.

Notice, however, that Doggett’s leapfrogging analysis—and c-command obviation analyses more generally—make a previously unnoticed set of related predictions: the properties of the pivot that are explained as side effects of the intermediate movement steps that enable an underlyingly *lower* PP to raise over a pivot should be optional—if not absent altogether—when an underlyingly *higher* PP raises to spec,TP. That is, in a

high PP/low DP configuration, raising the PP should be possible even if the DP pivot is prosodically light, unfocused, and pronounced in situ.¹¹

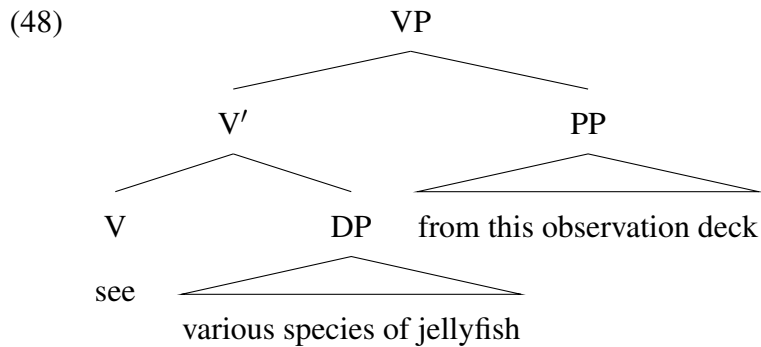
To test these predictions, let us start with simple cases involving a DP internal argument of a transitive verb and a PP modifier of the event described by the verb phrase; standard assumptions motivate an analysis in which the DP is the complement of the verb and the PP is an adjunct to the verb phrase, crucially above the DP. Consider the instance of LocInv in (47), for instance, where a *from*-phrase raises. That the PP is not a modifier of the DP is indicated by the fact that it is infelicitous to predicate the PP of the DP; rather, the PP modifies the meaning of the verb phrase by specifying the vantage point of the seeing event. We then have a schematic structure like that in (48), where the PP is underlyingly higher than the DP; this is corroborated by the ability of the PP to be stranded in *do so* substitution (49). A c-command obviation analysis then predicts that the PP may simply raise directly to spec,TP without focus movement of the DP. We see in (50b) that this prediction is not borne out. As in canonical instances of LocInv, it is ungrammatical for the pivot to be prosodically light and unfocused, unlike, for instance, the corresponding active sentence in (50a), even though it contains a superficially similar operation of PP topicalization.

(47) *Context:* A tour guide at an aquarium is describing how to view some jellyfish that are hidden behind some seaweed.

From this observation deck can be seen various species of jellyfish.

(cf. #Various species of jellyfish are from this observation deck).

¹¹Doggett (2004) assumes that raising of the DP to spec,vP is independently required by the strong phase impenetrability condition of Chomsky (2000), lest it not receive nominative case from T. But evidence from *there*-sentences and long passives indicates that the vP phase is accessible to probes on T for movement and agreement (Citko, 2014; Sheehan & Cyrino, 2023). I therefore assume that the pivot could in principle remain in VP and be case-licensed by T.



(49) I saw the jellyfish *from the hallway*, and you did so *from the observation deck*.

(50) (a) From this observation deck, you can see them. (PPTop)

(b) *From this observation deck can be seen them. (LocInv)

Next, to test whether high PP/low DP configurations require the DP pivot to be linearized to the right of the VP, let us consider structures in which the PP is an adjunct to a verb that embeds a clause containing the pivot. Consider the ECM construction in (51), where the locative PP is adjoined to a VP embedding an infinitival clause in which the pivot is the subject, as represented in (52).¹² This case forms a paradigm parallel to those considered in the previous section. That is, we find that transitive clauses allow the DP pivot to be linearized between the embedding verb and the embedded predicate (51a), whereas this word order is sharply degraded when the PP is fronted in LocInv (51b). Fronting of the PP instead requires the pivot to be prosodically heavy and pronounced to the right of the embedded predicate (51c), as in typical cases of LocInv.

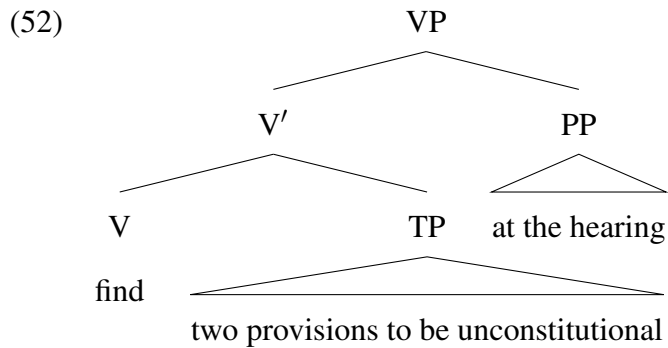
(51) *Context:* Yesterday, a court considered a constitutional challenge to a state law.

(a) The court found two provisions to be unconstitutional at the hearing.

(b) *At the hearing were found two provisions to be unconstitutional.

¹²For the sake of simplicity, I represent the embedded subject as remaining in the embedded clause when in immediate postverbal position. The force of the observations in this section is not diminished if one instead analyzes the subject as raising to the object position of the matrix verb phrase.

- (c) At the hearing were found to be unconstitutional two provisions restricting access to public records.



As with the examples in the previous section, the pronunciation of (51c) typically involves at least two prominent accents: one on the secondary predicate and another on the postposed pivot (53). A similar word order and intonation are independently observed in instances of HNPS like that in (54), where the pivot may be pronounced to the right of the in-situ locative PP, which Postal (1974) argued to involve movement of the pivot into the right-periphery of the matrix clause. In support of this hypothesis, consider the examples in (55) and (56), where we see that LocInv involves ordering the pivot to the right of the underlined matrix adverbials.

- (53) (At the hearing were found to be unconstiTUTIONal) || (two proVIsions restricting access to public records)
- (54) (The court found to be unconstiTUTIONal at the hearing) || (two proVIsions restricting access to public records).
- (55) (a) Professor Kunen proved three conjectures to be false (in the lecture) for the benefit of the students.
- (b) *In the lecture were proved three conjectures to be false for the benefit of the students.

- (c) In the lecture were proved to be false for the benefit of the students three conjectures that many people believe to be true.
- (56) (a) The journalist revealed several rumors to be true (in the report) after months of investigation.
- (b) **In the report* were revealed several rumors to be true after months of investigation.
- (c) In the report were revealed to be true after months of investigation several rumors that had previously been thought unlikely.

The above examples involve embedded infinitival clauses, but the same contrasts are found if the verb is independently allowed to take a small clause as its complement, as is the case for *find* and *prove*, but not *reveal*.

- (57) (a) At the hearing were found ⟨*two provisions⟩ unconstitutional ⟨two provisions restricting access to public records⟩.
- (cf. *The court found two provisions unconstitutional at the hearing.*)
- (b) In the lecture were proved ⟨*three conjectures⟩ false for the benefit of the students ⟨three conjectures that many people believe to be true⟩.
- (cf. *Professor Kunen proved three conjectures false in the lecture.*)

There is converging semantic and syntactic evidence for our assumption that the locative PP in each of these examples is a modifier of the matrix verb phrase, c-commanding the embedded DP pivot, as represented in (52). To use (55) as an example, the embedded predicate *false* is individual-level; hence we expect it to be infelicitous with locative modification (cf. Kratzer, 1995). This is what we indeed find when the verb is omitted (58), indicating that the locative in (55c) is necessarily modifying the event of proving, instead of somehow modifying the state of being false. Assuming that modifi-

cation approximately corresponds to syntactic sisterhood, this is evidence that the PP is merged not as an adjunct to the embedded small clause, but to the verb phrase.

(58) (a) The twin prime conjecture was proved (to be) false in the lecture.

(b) #The twin prime conjecture was false in the lecture.

A test from clausal topicalization confirms such a constituent structure; consider the paradigm in (59), where the embedded clause is finite. (Present tense is used to avoid confounds related to sequence-of-tense phenomena). Example (59a) is our baseline sentence; the question is whether the PP is adjoined to the embedded finite clause or to the matrix clause. In (59b), we see that the embedded clause may be topicalized, stranding the PP at the right edge of the sentence. If the PP could be base-generated within the embedded clause, we would expect that topicalization could carry the PP along with it, as is possible in cases like (60) involving a stage-level adjective (e.g., *visible*). We see that this expectation is not met in (59c), which is only grammatical under an odd reading in which falsity is somehow contingent.

(59) (a) Professor Kunen proves (that) the conjecture is false in this lecture.

(b) The conjecture is false, Professor Kunen proves in this lecture.

(c) #The conjecture is false in this lecture, Professor Kunen proves.

(60) The artifact is visible in this exhibit, I hear.

What we find, then, is that the focus, heaviness, and rightward linearization conditions on the pivot of LocInv hold even when the PP is base-generated higher than the pivot. Locality obviation analyses that assume that closeness is determined by c-command therefore overgenerate by incorrectly predicting that the pivot could remain unfocused, prosodically light, and in situ in such configurations.

In order to retain the explanatory force of locality obviation analyses in high

DP/low PP cases while expanding their empirical coverage also to high PP/low DP cases, I suggest that the relation central to locality is *precedence*, rather than c-command. The principles of linearization in English ordinarily conspire to yield a vP-internal word order in which DPs precede PPs, regardless of their underlying c-command relations. If so, then we can understand pivot postposing as a mechanism to prevent the pivot from linearly intervening between the PP and T, as illustrated in (61). When postposing does not apply, then a DP will precede a PP, even if the PP asymmetrically c-commands the DP; then the DP may raise (61a), but the PP may not (61b). However, if the DP is postposed, then it is linearized to the right of the PP and no longer intervenes for raising of PP (61c). If the mechanism for postposing is assimilated to the mechanism for HNPS, then the focus and prosodic heaviness conditions follow. This would, in effect, be an analysis in the spirit of Doggett (2004): we explain the conditions on the pivot as epiphenomena of independently motivated movements that allow a PP to obviate an intervening DP. This new formulation, however, would have the empirical advantage of applying to *all* cases of LocInv, not merely those in which the DP-PP c-command relations end up being reversed.

- (61) (a) *Three conjectures* were proved false in the lecture. (DP raised; PP in situ)

T ... [_{vP} ... prove [_{SC} [_{DP} *three conjectures*] false] ... [_{PP} in the lecture]]

- (b) **In the lecture* were proved *three conjectures* false. (PP raised; DP in situ)

T ... [_{vP} ... prove [_{SC} [_{DP} *three conjectures*] false] ... [_{PP} in the lecture]]

- (c) *In the lecture* were proved false *three conjectures* (that many people believe to be true). (PP raised; DP postposed)

T ... [_{vP} ... prove [_{SC} DP AP] ... PP ... DP ...]

4. Left-bracket Closeness

The empirical generalization established thus far is that raising a PP requires a competing DP to be postposed to the right of the PP. The remainder of this paper asks what notion of locality predicts this generalization. Independent motivation that locality is somehow sensitive to linear order comes from Hewett (2025), who presents data from pseudo-passives and multiple *wh*-questions in which neither of two potential goals c-commands the other, but where the goal on the left appears to intervene for movement of the other. In pseudo-passives, for example, Hewett observes that verbs like *speak* can freely order their two PP arguments (62ai, 62bi); but the (ii) and (iii) examples show that A-movement in pseudo-passivization may occur only out of the leftmost PP.¹³

- (62) (a) (i) The investigator spoke [PP with the witnesses] [PP about the crime scene]
(ii) [The witnesses]₁ were spoken [PP with —₁] [PP about the crime scene].
(iii) *[The crime scene]₁ was spoken [PP with the witnesses] [PP about —].
(b) (i) The investigator spoke [PP about the crime scene] [PP with the witnesses]
(ii) [The crime scene]₁ was spoken [PP about —] [PP with the witnesses].
(iii) *[The witnesses]₁ were spoken [PP about the crime scene] [PP with —].

There are strong and weak formulations of syntactic locality compatible with Hewett’s intervention effects. Hewett advocates for what I call a *strong* formulation of precedence-sensitive locality in which precedence directly replaces c-command in the formulation of “closest.” However, Hewett’s data are equally compatible with a *weak* formulation in which precedence is consulted only as a tie-breaker when c-command does not impose a unique choice among potential goals.¹⁴ I argue that the data from LocInv

¹³See Hewett (2025, 46ff.) for arguments that the effect in (62aiii) and (62biii) is not simply due to a requirement that the PP hosting the raised DP be adjacent to the verb.

¹⁴Such a tie-breaking mechanism could be implemented within Optimality Theoretic syntax, where

presented in the previous section provide evidence for a strong formulation, since even a c-commanding PP cannot raise unless its DP competitor first undergoes postposing.

To make this work, let us adopt minimal assumptions about the timing of linearization and probing. For Hewett’s data, it is sufficient to assume that constituents within the search domain of a probe are ordered; one way to implement this is for Merge to produce ordered pairs (Hewett, 2025, 8, fn.9). More needs to be said in the case of LocInv, however—consider again the contrast in (61). What we would like to say is that the *in situ* DP in (61b) precedes the PP and therefore intervenes for an Agree operation that would result in T attracting the PP; but (61c) is grammatical because postposing of the DP results in an intermediate linear order in which the PP precedes the DP, so that the DP does not intervene when T probes for a goal.

I assume that Merge is the sole structure-building mechanism in syntax, where movement of an XP is the consequence of the XP undergoing Merge with a YP that dominates the XP (“internal” Merge; Chomsky, 2004). Then movement produces multidominant structures (i.e., graphs) instead of trees, so that the DP and the PP in (61c) are not automatically ordered: even if Merge produces ordered pairs, this would yield a structure in which the DP both precedes and follows the PP.

To avoid this issue, I assume that linearization applies over the course of the derivation, generating a set of precedence statements among syntactic objects, as is assumed in work on Cyclic Linearization (Fox & Pesetsky, 2005). In particular, when a phrase has undergone overt movement, its highest occurrence is used for calculating the phrase’s order relative to other material in the tree; lower occurrences remain available for interpretation but not for linearization (except in the case of covert movement, which I ignore here). Unlike Cyclic Linearization, however, I am not committed to the premise

lower-ranked constraints determine the winner among candidates not distinguished by higher-ranked constraints; for an introduction to OT syntax, see the papers in Legendre et al. (2001).

that precedence statements are frozen at phase spell-out. It is also immaterial whether these statements are generated merge-by-merge or phase-by-phase, provided that clause-internal movement is reflected in the ordering that is available when T probes. What is essential for the present analysis is that, when a probe launches a search operation, the relevant precedence statements are used to construct a bracketed representation of its search domain.

I suggest that a probe’s search domain is ordered by the locality metric which I call *Left-bracket Closeness* (LBC; 63). This captures the core effects of Hewett’s (2025) procedural formulation of Agree, where a probe H searches for a goal in a top-down, left-to-right manner, such that, upon interacting with a node $[\alpha \beta \gamma]$, H exhaustively interacts with its left daughter β before exhaustively interacting with its right daughter γ —a pre-order depth-first search procedure (cf. Branan and Erlewine, 2022). For the purposes of this paper, I do not take a stance on the issue of whether Agree is best formulated procedurally or representationally. My claim is narrower: whatever implementation is chosen, Left-bracket Closeness underlies Attract Closest (1). The alternative locality metrics to be considered are C-command closeness augmented with precedence-sensitive tie-breaking (CCC; 64) and Path-quantification Closeness (PQC; 65; Fitzpatrick, 2002; Müller, 1998; Richards, 1997).

(63) *Left-bracket Closeness* (LBC; Strongly precedence-sensitive locality)

Given two accessible goals G and K in the search domain of a probe P, G is closer to P than K iff the left bracket of G’s linearized occurrence precedes the left bracket of K’s linearized occurrence in the bracketed representation of that domain.

	Pseudopassive; PP₁ precedes PP₂	LocInv; DP precedes and c-commands PP	LocInv; DP precedes PP but PP c-commands DP
C-command with tie-breaking (CCC)	Chooses comp, PP ₁	Blocks PP raising	Allows PP raising
Path-quantification (PQC)	Chooses comp, PP ₂ (assuming layered structure)	Blocks PP raising	Allows PP raising
Precedence (LBC)	Chooses comp, PP ₁	Blocks PP raising	<i>Blocks PP raising</i>

Table 1: Predictions of three locality metrics, where shaded cells indicate incorrect predictions. The decisive case is the final column: when PP c-commands DP but DP precedes PP, only Left-bracket Closeness predicts that DP intervenes for PP raising.

(64) *C-command Closeness with Tie-breaking (CCC; Weakly precedence-sensitive locality)*

Given two accessible goals G and K in the search domain of a probe P, G is closer to P than K if G asymmetrically c-commands K. If neither of G and K c-commands the other, then G is closer to P than K if G precedes K.

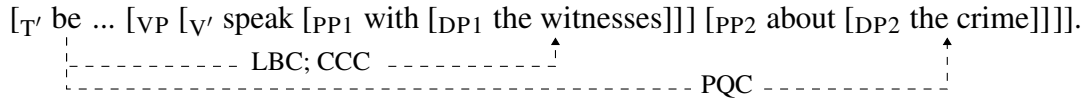
(65) *Path-quantification Closeness (PQC)*

Given two accessible goals G and K for a probe P, G is closer to P than K iff G has a shorter *path* to P than K does, where the path from α to β is the set of nodes γ that dominate α but not β .

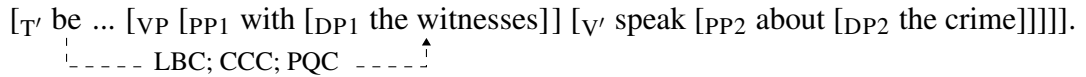
The predictions of these different locality metrics are given in Table 1. Before turning to the detailed analysis of LocInv, consider the two possible structures in (66) for the pseudopassive construction in (62), where I assume that the accessible goals are the DP complements of the two PPs. Under LBC, the hierarchical relation between the two PPs is irrelevant; what matters is the fact that DP₁ precedes DP₂, which makes DP₁ closer to T. Then Attract Closest allows only DP₁ to move, deriving the contrast between (62aii) and (62aiii). By comparison, the standard version of C-command Closeness without incorporating precedence-sensitive tie-breaking does not determine which of DP₁ or DP₂ is

closer to T, since the two DPs are not in a c-command relation. The tie-breaking version CCC makes DP₁ closer to T. The hierarchical relation between the two PPs becomes relevant only under PQC. To use the terminology of Pesetsky (1995), if PP₂ c-commands PP₁ in a “layered” structure, then DP₂ has a shorter path to T (66a); if PP₁ c-commands PP₂ in a “cascade” structure, then DP₁ has a shorter path to T (66b). Hewett (2025) argues that the layered structure is correct, in which case only LBC and CCC make the correct predictions.

(66) (a) *Predictions for locality in a layered structure*



(b) *Predictions for locality in a cascade structure*



The “accessible” proviso in (63-65) is intended to capture the effects of the Phase Impenetrability Condition and related formulations of subjacency while being agnostic as to their exact formulation (cf. Citko, 2014 for a review of various approaches). In the case of Left-bracket Closeness, this allows us to avoid incorrect predictions of naïve, structure-independent formulations of precedence-based locality (cf. Rizzi, 2013). For instance, a *whP* embedded in a [-*wh*] subject does not (always) intervene for movement of a *whP* object in a sentence like (67).¹⁵ But *where* is plausibly within the NP spell-out domain of the DP phase (shaded), hence not accessible to C, which has as its sole accessible goal the *whP* object *what*. Under certain conditions, an embedded phrase may become phase-

¹⁵The existence of intervention effects in cases like (67) is controversial. For further discussion, see Heck and Müller (2003) and Müller (2011). The point here is that the non-existence of intervention effects in (67) does not automatically disqualify a theory of syntactic intervention in which the central relation is precedence instead of c-command.

accessible to an unembedded probe by moving to the edge of the phase that immediately dominates it, but it does not obligatorily do so, provided that it may be licensed in-situ, as in multiple *wh*-questions.

- (67) What₁ did [_{DP} D_Ø [_{NP} people from *where*]] try to buy —₁? (Fiengo et al., 1988, 89, ex.20)

By comparison, consider Hewett’s multiple PP constructions (62). Abels (2003) argues that PPs must not be phases in English, since P-stranding is possible, and an antilocality condition prohibits the movement of the comp,PP to spec,PP that would be necessary if PP were a phase. If so, then DP₁ and DP₂ are both accessible to the agreement probe on T in (62).

5. Deriving the alternation between inverted and non-inverted clauses

5.1. The locality puzzle revisited

Recall the puzzle posed in Section 2 for the alternation between inverted and non-inverted clauses. For a given assignment of theta roles to a DP and a locative PP, either of the potential hierarchical relations between the DP and the PP in (68a) and (68b) would seem to involve non-local movement. If the PP is underlyingly higher than the DP (68a), then raising of the DP would violate C-command closeness or Path-quantification closeness.¹⁶ If, on the other hand, the DP is higher than the PP (68b; cf. Section 2.2), then raising the PP in LocInv would violate any of the locality metrics under consideration. What the pivot postposing condition on LocInv (26) tells us is that the movement in (68b) is, in fact, ungrammatical. Raising of the PP must instead involve postposing of the DP pivot, represented in (68c) as movement of the DP to a right specifier of vP. (Rightward

¹⁶If the PP is a *by*-phrase, then this resembles the puzzle of raising an internal argument over an external argument in a passive construction (cf. Collins, 2005).

movement is not essential to the analysis; alternatives are considered below).

- (68) (a) [TP DP [T' T ... [VP ... [... DP ...] ... PP ...]]

 (b) *[TP PP [T' T ... [VP ... DP ... [... PP ...]]]

 (c) [TP PP [T' T ... [vP [VP ... DP ... [... PP ...]] DP]]


I argue that Left-bracket Closeness best explains the paradigm in (68). As representative cases, let us derive (69) and (70), where the contrast between the (b) and (c) cases illustrates the pivot postposing condition. The central claim is that Left-bracket Closeness is able to explain why this contrast occurs both when the DP underlyingly c-commands the PP (69) *and* when the opposite holds (70), whereas the other locality metrics—including the “weak” tie-breaking version of precedence-sensitive locality—can only explain the former case (cf. Table 1).

- (69) (a) Various medical supplies are made available to the students. (high DP raised; low PP in situ)
 (b) *To the students are made various medical supplies available. (low PP raised; high DP in situ)
 (c) To the students are made available various medical supplies. (low PP raised; high DP postposed)
- (70) (a) Three conjectures were proved false in the lecture. (low DP raised; high PP in situ)
 (b) *In the lecture were proved three conjectures false. (high PP raised; low DP in situ)

- (c) In the lecture were proved false three conjectures. (high PP raised; low DP postposed)

5.2. Mechanical assumptions

My assumptions about the mechanics of agreement and EPP-checking are as follows. What we would like is for T to attract the closest movable goal XP and to agree with the closest number-bearing goal YP—XP, if possible—otherwise the next closest number-bearing goal. In the case of ordinary DP raising in English (e.g., 69a), the XP and YP are both the same DP, but in LocInv, the XP may be a PP and the YP may be a DP.¹⁷ Here, I propose one set of assumptions that allows us to achieve this outcome, acknowledging that they are not essential to our analysis of locality.

I suggest that the agreement probe on T in English is specified as [INT: Φ , SAT: #], formulated within the interaction/satisfaction framework for Agree of Deal (2015) and subsequent work. Thus, the potential goals of T are those phrases in its search domain which bear at least a minimal ϕ -feature specification; T copies the features from (“interacts” with) these potential goals until it copies a number feature, when T is “satisfied” and halts. The order in which T interacts with its goals is given by Left-bracket Closeness. Though not the focus of our analysis, I assume that T may license a DP with which it has interacted, which is sometimes realized as nominative case.¹⁸

As for the relation between agreement and movement, I follow Longenbaugh (2019) in implementing the EPP as a subcategorization feature on T; I assume that this is

¹⁷XP and YP also differ in expletive-subject constructions and in various inversion constructions other than LocInv (e.g., vP predicate inversion; cf. Samko, 2016), the analysis of which is beyond the scope of this paper.

¹⁸In LocInv, Iwakura (1978, p. 329) observes that the *wh*-pivot of an echo question must be nominative (i). For personal pronouns, however, there is a preference for realizing the pivot as accusative (cf. 73). The analysis of this difference is beyond the scope of this paper.

- (i) A. Among the guests sat the manager of a large hedge fund.
B. Among the guests sat *who*(**m*)?

[• ϕ •], which licenses a ϕ -bearing XP to undergo Merge with a projection of T. In order to ensure that this XP is the goal closest to T, I recast Longenbaugh's economy principle of Feature Maximality within the interaction/satisfaction framework (71).

(71) *Feature Maximality*

If, at some stage, XP discharges any requirement of H (by interaction, satisfaction, or Merge), then XP must also discharge every other requirement of H that XP can discharge at that stage.

I adopt the following assumptions about the features borne by DPs and PPs. All DPs bear at least a set of ϕ -features that includes number; thus, they always intervene for the raising of another ϕ -bearing XP in the relevant configuration. The fact that locative PPs may raise suggests that they too bear certain ϕ -features.¹⁹ I analyze locative PPs as bearing a minimal referential feature [π], which is the root of the feature geometry for person and deixis (approximately [R] in the system of Bejar, 2003). Thus, PPs that participate in LocInv are able to introduce spatial discourse referents that can be anaphorically referred to by the locative proform *there* (72); deictic locatives such as *here* and *there* are evaluated relative to a discourse center; and LocInv bleeds person agreement with a pivot (73). For the sake of compactness, the subscripts on a given DP or PP in subsequent derivations should be understood as representing only a relevant subset of their ϕ -features.

(72) [Into the room]_i walked Robin. He found the Joker there_i. (Doner, 2019, 103, ex. 31)

(73) In this picture is/*am me, and in that one is/*are you. (Longenbaugh, 2019, 65, ex. 6a)

¹⁹The assumption that what moves to spec,TP in LocInv is a PP and not a null expletive is corroborated in Section 6.

5.3. Derivations without postposing of the pivot

Given these assumptions, let us derive the contrasts in (69a-b), which do not involve postposing of the pivot. Consider first the ordinary passive (74). I assume that *make* takes a small-clause complement whose subject is a DP and whose predicate is an AP; here, A^0 in turn takes a PP complement (I)—cf. (17). Under any locality metric considered here, the DP is identified as the closest goal to T, so that T copies its ϕ -features (II). Because DP bears a plural number feature, T meets its satisfaction condition and does not interact with any further goal. And, by Feature Maximality, the DP must also discharge the subcategorization feature of T by Merge in spec,TP (III). Thus, the agreement with the PP necessary to raise it and reverse the underlying DP-PP c-command relations is ungrammatical (75). This yields the ordinary passive sentence in (69a) in which the DP is pronounced in spec,TP and controls subject agreement.

- (74) *Target sentence*: Various medical supplies are made available to the students. (✓
LBC; ✓ CCC; ✓ PQC)

(I) $[_{VP} be_{pass} [_{VP} make [_{SC} DP_{[PL]} [_{AP} A^0 PP_{[\pi]}]]]]$

(II) $[_{T'} T_{[\bullet\Phi, INT: \Phi, SAT: \#, PL]} [_{VP} be_{pass} [_{VP} make [_{SC} DP_{[PL]} [_{AP} A^0 PP_{[\pi]}]]]]]$
----- ✓ LBC ; ✓ CCC ; ✓ PQC -----

(III) $[_{TP} DP_{[PL, NOM]} [_{T'} T_{[\bullet\Phi, PL]} [_{VP} be_{pass} [_{VP} V^0 [_{SC} DP [_{AP} A^0 PP_{[\pi]}]]]]]]]$
↑

- (75) *Target sentence*: *To the students are made various medical supplies available. (✗
LBC; ✗ CCC; ✗ PQC)

(I) $[_{VP} be_{pass} [_{VP} make [_{SC} DP_{[PL]} [_{AP} A^0 PP_{[\pi]}]]]]$

(II) $[_{T'} T_{[\bullet\Phi, INT: \Phi, SAT: \#, PL]} [_{VP} be_{pass} [_{VP} make [_{SC} DP_{[PL]} [_{AP} A^0 PP_{[\pi]}]]]]]$
----- ✗ LBC; ✗ CCC; ✗ PQC -----

Each of the three locality metrics under consideration correctly generates the contrast between (69a) and (69b), as summarized in the second column of Table 1. The

predictions of LBC diverge, however, when the PP is base-generated above the DP as in (70); consider the derivation in (76). In contrast to (69), the PP in (70) is an adjunct to VP, crucially *above* the DP (I; cf. Section 3.3). Next, T enters the derivation (II). The PP c-commands the DP, so the PP is deemed closest to T under CCC and PQC. By Feature Maximality, PP must also check T's subcategorization feature, forcing it to undergo internal Merge with T' (III). T's satisfaction condition is not met, so it continues probing, interacting with the DP as its next-closest goal (IV). Thus, T copies the features of DP and assigns it nominative case. The resulting sentence is only predicted to be ungrammatical by LBC (cf. the third column of Table 1).

(76) *Target sentence*: *In the lecture were proved three conjectures false. (✗ LBC; ✓ CCC; ✓ PQC)

(I) [_{VP} be_{pass} [_{VP} [_{V'} prove [_{SC} DP_[PL] false]] PP_[π]]]

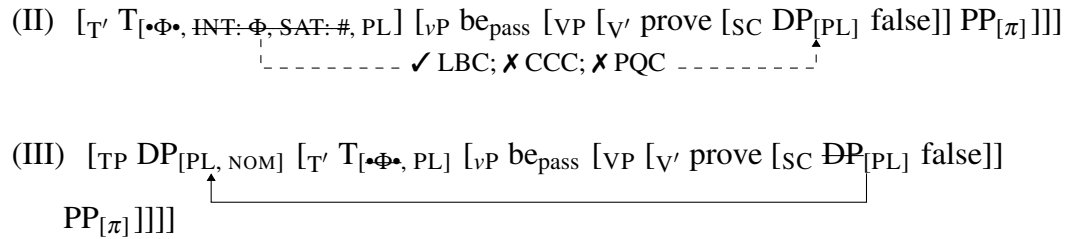
(II) [_{T'} T_[•Φ, INT: Φ, SAT: #] [_{VP} be_{pass} [_{VP} [_{V'} prove [_{SC} DP_[PL] false]] PP_[π]]]]

(III) [_{TP} PP [_{T'} T_[•Φ, INT: Φ, SAT: #] [_{VP} be [_{VP} [_{V'} prove [_{SC} DP_[PL] AP]] PP_[π]]]]]

(IV) [_{TP} PP [_{T'} T_[INT: Φ, SAT: #, PL] [_{VP} be [_{VP} [_{V'} prove [_{SC} DP_[PL, NOM] AP]] PP_[π]]]]]

Left-bracket Closeness avoids this undesired outcome, because stage (II) would involve T first interacting with the DP, deriving an ordinary passive sentence (77). In contrast, CCC and PQC incorrectly predict the PP to intervene for raising of the DP. To summarize, the locality metrics other than LBC predict what raises in low DP/high PP configurations to be the *opposite* of what is actually found.

(77) *Target sentence*: Three conjectures were proved false in the lecture. (✓ LBC; ✗ CCC; ✗ PQC)



5.4. Derivations with postposing of the pivot

To complete the analysis of (69) and (70), let us consider how postposing of the DP pivot enables the PP to raise, beginning by more carefully examining the mechanics of postposing.

A lower bound for the structural position of the postposed DP is provided by adjectival secondary predicates, which indicate that the pivot is pronounced at least as high as VP (cf. Section 3). NPI licensing provides an upper bound: cases like (78) and (79) show that an NPI pivot may be licensed by sentential negation and *never*, both of which are structurally lower than T. Therefore, the postposed pivot of LocInv lies structurally between VP and T; let us assume that this position is spec,vP (Doggett, 2004).

- (78) In the kitchen must *(not) be kept any flammable material.
- (79) My client maintains that in his vehicle had *(never) been stored any illicit substances.

We have seen in Section 3 that pivot postposing in LocInv shares with HNPS the condition that the pivot must be focused. Based on these parallels, Doggett (2004, 2005) analyzes the pivot as being attracted by a [uFoc] probe on v to a right specifier of vP .²⁰ Though not observed by Doggett, pivot postposing in LocInv is additionally like HNPS in

²⁰ Although the direction of attachment is stipulated within the account of Doggett (2004, 2005), Ernst's (2004) treatment of HNPS offers a more principled explanation within his Weight Theory: in a nutshell, "heavy" constituents are linearized to the right within the extended projection of the verb—not only the DPs moved via HNPS, but PPs and CPs as well. None of this is essential for our present purposes.

licensing parasitic gaps (80). Moreover, Larson’s (1988) generalization holds: postposing makes the parasitic gap in adjuncts of the kind headed by *before* and *without* not only possible, but obligatory.

- (80) At the auction had been sold—before the provenance of *pg/??*them could be verified—a set of Cubist sketches purportedly by Picasso.

In the following derivations, I use these distributional facts as a warrant to represent pivot postposing as $\bar{A}$ -movement to a right-hand spec,vP, as illustrated in (81). It is not necessary to commit to a rightward movement analysis, however. There exists an alternative smuggling analysis (82), for instance, in which the DP pivot is first moved to a left-hand spec,vP, followed by remnant movement of the VP to a higher specifier (Storment, 2025).²¹ What is more important than the mechanism underlying postposing, however, is the question of why it must occur in high PP/low DP configurations like (70). We have already seen in (76) that the locality metrics other than LBC incorrectly permit the pivot to remain in situ.

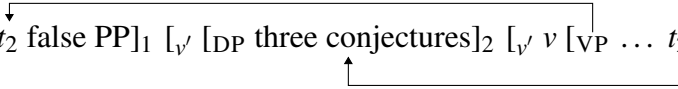
- (81) *Rightward movement analysis of pivot postposing*

[_{vP} [_{v'} *v* [_{VP} prove *t*₁ false PP]] [_{DP} three conjectures]₁]



- (82) *Smuggling analysis of pivot postposing*

[_{vP} [_{VP} prove *t*₂ false PP]₁ [_{v'} [_{DP} three conjectures]₂ [_{v'} *v* [_{VP} ... *t*₂ ...]₁]]]



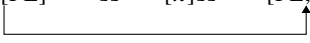
Now consider the derivation of (70c) in (83). The DP and the PP both enter the derivation with ϕ -features, but the DP additionally has a focus feature (I). Next, *v* enters the derivation with a probe specified to identify a single constituent bearing focus and attract it to its specifier. The DP is identified as its goal (II-III). Next, when T probes,

²¹See Overfelt (2015) for discussion of the similar debate over the direction of movement in HNPS.

Left-bracket Closeness then requires it to first interact with the PP (IV). Without additional assumptions, CCC and PQC predict the postposed DP to intervene for agreement between T and the PP. By Feature Maximality, T must also use the PP to check its subcategorization feature (V). Because T has not yet checked its satisfaction feature, it continues probing and identifies DP as the next closest goal (VI); as a result, it copies its number features and assigns nominative case to the DP.

- (83) *Target sentence*: In the lecture were proved false three conjectures (that are widely believed to be true). (✓ LBC; ✗ CCC; ✗ PQC)

- (I) $[_{VP} [_{V'} \text{prove} [_{SC} \text{DP}_{[PL, FOC]} \text{false}]] \text{PP}_{[\pi]}]$
- (II) $[_{V'} \text{be}_{\text{pass}[\bullet \text{FOC} \bullet, \text{INT}: \text{FOC}, \text{SAT}: \text{FOC}]} [_{VP} [_{V'} V [_{SC} \text{DP}_{[PL, FOC]} \text{AP}]] \text{PP}_{[\pi]}]]]$

- (III) $[_{vP} [_{V'} \text{be}_{[\bullet \text{FOC} \bullet]} [_{VP} [_{V'} V [_{SC} \text{DP}_{[PL]} \text{AP}]] \text{PP}_{[\pi]}]] \text{DP}_{[PL, FOC]}]$

- (IV) $[_{T'} T_{[\bullet \Phi \bullet, \text{INT}: \Phi, \text{SAT}: \#]} [_{vP} [_{V'} v^0 [_{VP} [_{V'} V [_{SC} \text{DP}_{[PL]} \text{AP}]] \text{PP}_{[\pi]}]] \text{DP}_{[PL, FOC]}]]]$

- (V) $[_{TP} \text{PP}_{[\pi]} [_{T'} T_{[\bullet \Phi \bullet, \text{INT}: \Phi, \text{SAT}: \#]} [_{vP} [_{V'} v [_{VP} [_{V'} V [_{SC} \text{DP} \text{AP}]] \text{PP}_{[\pi]}]] \text{DP}_{[PL, FOC]}]]]]]$

- (VI) $[_{TP} \text{PP} [_{T'} T_{[\text{INT}: \Phi, \text{SAT}: \#, \text{PL}]} [_{vP} [_{V'} v [_{VP} [_{V'} V [_{SC} \text{DP} \text{AP}]] \text{PP}]] \text{DP}_{[PL, \text{NOM}]}]]]]]$


Thus, we find that LBC allows us to understand postposing as a syntactic demotion process, enabled by an independently available $\bar{A}$ -probe on v . The focus condition on the pivot is then explained as a side effect of this probe, much as in the account of Doggett (2004). Unlike previous locality obviation analyses of pivot postposing, however, the present account extends to *all* cases of PP raising, not merely those involving high DP/low PP configurations.

6. Subextraction from the pivot: Locative inversion versus PP topicalization

Our analysis of LocInv contains two crucial claims: (i) syntactic locality is determined by Left-bracket Closeness, and, more narrowly, (ii) movement of a PP in LocInv patterns with A-movement of a DP for syntactic intervention. Data presented in the preceding sections involved two accessible goals in a precedence relation. But LBC also makes predictions when one goal contains the other, since a container’s left bracket precedes the left bracket of the contained phrase. Authors including Bresnan (1976), Fukui (2006), Müller (1996), and Sauerland (1999) have argued that such configurations obey a version of what Müller (2011) terms the “F-over-F principle,” formulated in (84) as a condition, rather than a principle. Whereas the classical A-over-A principle of Chomsky (1964) bars subextraction of one constituent from another of the same category, the F-over-F Condition is instead relativized to the feature [F] of a given probe.

(84) *F-over-F Condition*

The structure $[H_{[F, INT: F]} \dots [Y_{P[F]} \dots X_{P[F]} \dots] \dots]$ may not involve Agree between H and XP, where YP dominates XP and bears an active [F] feature.²²

Fukui (2006) and Sauerland (1999) use versions of this condition to explain contrasts like that in (85), which were originally analyzed by Saito (1992) in terms of the Proper Binding Condition. The so-called “surfing” derivation in (85a) involves movement of whP_2 to the embedded spec,CP, followed by subextraction of whP_1 to matrix spec,CP; this second movement step is understood not to violate (84) because whP_2 has already satisfied the embedded [wh]-probe and is therefore no longer an active goal for the matrix [wh]-probe. In contrast, the “diving” derivation of (85b) involves subextraction of whP_1 to

²²I use *active* here in a weak sense: a phrase is active for a probe iff it is available as a possible goal for that probe at the relevant derivational point. In order to analyze the surfing/diving contrast, I assume that a whP is rendered inactive for subsequent wh -probes when it has undergone wh -movement.

the embedded spec,CP as its initial step, violating the F-over-F Condition.²³

- (85) (a) ?[What film]₁ did Kim ask [what part of ₁]₂ to watch ₂? (Surfing)
-
- (b) *[What part of ₁]₂ did Kim ask [what film]₁ to watch [what part of ₁]₂? (Diving)
-

The F-over-F Condition is sometimes derived as a special case of Path-quantification Closeness (see Fitzpatrick, 2002 for further discussion). However, PQC is empirically inadequate for reasons discussed in Section 5 and summarized in Table 1. Instead, I suggest that the F-over-F Condition is best viewed as a special case of Left-bracket Closeness: if YP dominates XP, the left bracket of YP typically precedes the left bracket of XP.²⁴ Thus, the same search metric that treats a preceding goal as closer than one that follows it also treats a dominating goal as closer than one that is contained within it. F-over-F effects therefore need not be imposed by the separate domination-sensitive condition in (84); they follow from a general ordering of goals intuitively induced by top-down, left-to-right search.

Let us consider the role of the F-over-F Condition in LocInv, which has not been

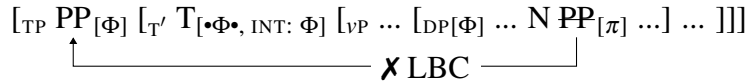
²³The surfing/diving contrast is a case of the Müller-Takano generalization, according to which α -subextraction of XP out of YP followed by α -movement of YP is less acceptable than α -movement of YP followed by α -subextraction of XP out of YP, where α ranges over scrambling, *wh*-movement, etc. (e.g., Kitahara, 1994; Müller, 1996; Takano, 1994).

²⁴This would only fail to be the case in a remnant movement derivation in which XP is first subextracted from YP and then YP undergoes rightward movement to a position above XP. LocInv via subextraction is in fact marginally acceptable in such configurations. Notice that postposing of the pivot in (i) makes subextraction marginally acceptable in LocInv. HNPS in presentational *there* sentences like (ii) suggests that it is grammatical for a PP to undergo extraposition out of the DP, followed by postposing of the DP, in a remnant movement derivation. In the absence of *there*, these two movements would result in a ν P-internal word order in which the PP precedes the DP within which it was base-generated, making the PP closest to T under LBC.

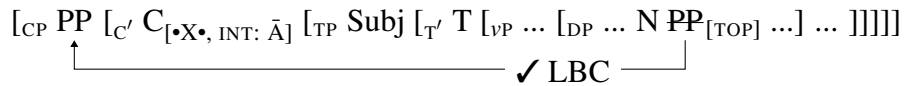
- (i) [*To this problem*]₁ has been $\langle$ *[a possible solution t_1] $\rangle$ discovered $\langle$?[a possible solution t_1] $\rangle$.
- (ii) ?There has been discovered t_2 [*to this problem*]₁—[a possible solution t_1]₂.

investigated previously, to my knowledge. If movement of a PP in LocInv is of the same kind as ordinary A-movement of a DP, as was assumed in Section 5, and if Left-bracket Closeness is a general locality principle, then we expect that a PP may *not* cross a dominating DP in LocInv (86), because DP is a closer goal to T. I assume that a PP is base-generated within DP in cases where the head noun N^0 l-selects for the preposition P^0 that heads PP (e.g., *dependence* + *on* DP). But $\bar{A}$ -movement of PP (say, in PP topicalization) should not be blocked by the presence of a dominating DP (87), because probes for $\bar{A}$ -movement in English are insensitive to the presence of DP interveners that lack $\bar{A}$ -features (provided a mechanism for obviating subjacency in subextraction, such as successive-cyclic movement through spec,DP). Let us assume, concretely, that C may bear a probe of the form $[\bullet X \bullet, \text{INT: } \bar{A}]$, attracting the closest phrase of any syntactic category that bears an $\bar{A}$ -feature, where $[\bar{A}]$ ranges over [TOP], [FOC], [*wh*], and possibly other features.²⁵

(86) *LBC predicts a dominating DP to intervene for A-movement of a PP*



(87) *LBC predicts a dominating DP not to intervene for $\bar{A}$ -movement of a PP*



There are at least two ways for these predictions to turn out to be incorrect. First, it could be that the F-over-F Condition is incorrect, so that subextraction in (86) is actually grammatical; if so, then Left-bracket Closeness would not be correct. Alternatively, it could be that the PP of LocInv is not, in fact, moving to subject position, or that the attracting head is not sensitive to the presence of intervening DPs. This would be the case

²⁵The use of an $[\bullet X \bullet]$ subcategorization feature is borrowed from Longenbaugh (2019) and Newman (2024).

under the analyses of Bruening (2010), Coopmans (1989), and Postal (2004), who argue that LocInv involves topicalization-like $\bar{A}$ -movement of a PP, where the actual subject is a silent counterpart of expletive *there*. Such an analysis involves three main components: (i) a null expletive *there*_∅ is inserted in spec,TP; (ii) a PP merges above *there*_∅, say, in spec,CP; and (iii) there is a licensing condition such that (i) is only grammatical if (ii) also occurs.²⁶ If so, then LocInv with a covert expletive subject should be grammatical when PP topicalization with an overt expletive subject is grammatical, and vice versa.

The paradigm in (88) bears out the predictions of (86) and (87). When a PP is base-generated within a VP, it may move either via topicalization (88a) or via LocInv (88b). However, when the nominal counterpart of the same verb selects for the PP, there emerges a strong contrast: PP topicalization is acceptable (88c), but PP movement in LocInv is sharply ungrammatical (88d). This paradigm is quite general, as illustrated in (89-90).²⁷

- (88) (a) From the final score, there were deducted several points. (PPTop; PP in VP)
 (b) From the final score were deducted several points. (LocInv; PP in VP)
 (c) From the final score, there were several deductions still to be made. (PPTop; PP in NP)
 (d) *From the final score were several deductions still to be made. (LocInv; PP in NP)

²⁶For a concrete formulation of the licensing rule, Bruening (2010, p. 79) proposes that “an expletive in Spec-IP can only be null when it is associated with a fronted phrase (adjoined to IP or moved to Spec-CP).” The relation of this rule to other grammatical phenomena is not discussed.

²⁷As a sanity check that PP indeed moves from a position local to the selecting N, note that subextraction is sensitive to islands dominating DP, such as the adjunct island in (i), whereas the baseline sentence in (ii) is acceptable.

- (i) *[From the final score]₁, the team got angry [because there were [too many deductions —₁]].
 (ii) The team got angry [because there were [too many deductions [from the final score]]].

- (89) (a) To the new fund, there has already been donated a substantial sum of money.
 (b) To the new fund has already been donated a substantial sum of money.
 (c) To the new fund, there is already a substantial donation scheduled for this month.
 (d) *To the new fund is already a substantial donation scheduled for this month.
- (90) (a) On this technology, there depend countless lives.
 (b) On this technology depend countless lives.
 (c) On this technology, there is widespread dependence.
 (d) *On this technology is widespread dependence.

A detailed study of expletive-subject sentences is beyond the scope of this paper; for the sake of concreteness, however, let us follow Francez (2007) in analyzing the DP pivot of an existential sentence like (88c) as a generalized quantifier functioning as the clause's main predicate, with optional codas being adjuncts to VP. Then topicalization from out of the pivot is a familiar and theoretically unproblematic case of $\bar{A}$ -movement of a complement (91) (The optional coda is omitted for the sake of space). If the null expletive analysis of locative inversion were correct, then, the structurally identical structure formed by replacing *there* with its null counterpart *there*_Ø would be incorrectly predicted to be grammatical (92).

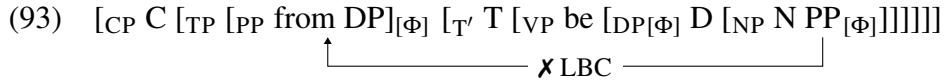
- (91) From the final score, there were several deductions (still to be made).

[CP [PP from DP]_[TOP] [C' C_[•TOP•] [TP *there* [T' T [VP be [DP[Φ] D [NP N PP_[TOP]]]]]]]]

- (92) *From the final score were several deductions (still to be made).

[CP [PP from DP]_[TOP] [C' C_[•TOP•] [TP *there*_Ø [T' T [VP be [DP[Φ] D [NP N PP_[TOP]]]]]]]]

The PP raising/PP-as-subject analysis, on the other hand, gives us a straightforward way to understand the contrast between (88c) and (88d). Rather than positing a null expletive subject, we analyze LocInv as involving A-movement of PP to spec,TP, triggered by ϕ -Agree. Then subextraction from the pivot via LocInv is a violation of Left-bracket Closeness, because the left bracket of the DP precedes the left bracket of the PP that DP dominates (93).²⁸



7. Conclusion and broader implications

I have argued that postposing of the pivot in English LocInv may be understood as a syntactic demotion process: a preceding, closer goal to a probe may be rendered more distant by being postposed to the right of another goal. The analysis depends on three assumptions: that the PP of LocInv undergoes A-movement or at least movement subject to A-locality; that the pivot must be postposed in LocInv; and that the search domain of T is ordered according to Left-bracket Closeness. The first two have been argued for empirically; the third is the theoretical conclusion drawn from the data.

This analysis suggests a broader typological hypothesis: some inversion constructions are made possible by (ideally, independently licensed) postposing operations that alter the order of accessible goals. I mention a potential application to OVS sentences here. Ndayiragije (1999) has argued that subject-object reversal in Kirundi requires postposing

²⁸The reader may wonder whether the pivot itself may raise, given that it is closest to T under LBC. What we find is that this is possible when a coda is available to function as the predicate of the raised DP in the resulting copular sentence, as indicated by (i-ii), but not when a coda is absent (iii). The analysis of the ungrammaticality of the latter case is beyond the scope of this paper.

- (i) Several deductions from the final score were still to be made.
- (ii) A substantial donation to the new fund is already scheduled for this month.
- (iii) *Widespread dependence on this technology is.

of the subject to a VP-external right-hand focus position, as with our analysis of LocInv, citing minimal pairs like that in (94). Here, we see that internal arguments ordinarily precede postverbal adverbs like *néézá* ‘well’ (94a), but the subject in an OVS sentence must follow them (94b), indicating that subjects in such sentences occupy a right-peripheral position. Similarly, Russian OVS has been analyzed as involving A-movement of the object accompanied by postposing of the subject (e.g., Antonyuk, 2021; Pereltsvaig, 2021). The theory of locality presented in this paper gives us a new way to understand how such constructions can be compatible with syntactic locality; postposing of the nominative subject to the right of the accusative object renders the object closer to T than the subject, by Left-bracket Closeness. The full range of inversion constructions derivable in this way remains to be seen.

- (94) (a) yohani a-á-ra-oógeje <imiduga> néézá <*imiduga>
 John 3S-PST-ra-wash cars well *cars
 ‘John washed cars well.’
- (b) imiduga yi-á-oógeje <*yohani> néézá <yohani_[+FOC/*-FOC]>
 cars 3S-PST-ra-wash *John well John
 ‘John washed cars well.’ (Ndayiragije, 1999, pp. 416–7)

Also important for further investigation are deep questions about the etiology of the cross-linguistic tendency to place presentational focus on the nominal pivot of inversion constructions (cf. Bresnan and Kanerva, 1989; Marten and Gibson, 2016). Potentially, this association originates in a soft information-structural alignment preference whereby clause-initial position favors topical or frame-setting material, while clause-final/postverbal position favors focal or less discourse-familiar material (e.g., Gundel, 1988). Nonetheless, the same information-structural preferences need not be syntactically enforced by the same means across languages. If the present paper is correct, postposing of a linearly intervening goal is one syntactic strategy for a probe to agree with an apparently non-local goal; this does not preclude inversion constructions elsewhere from

employing different strategies to the same effect.

Zooming out further, this paper adds to a growing body of work that assigns a central role to precedence in various components of the grammar, including variable binding (Barker, 2012), Condition C of the binding theory (Bruening, 2014), and presupposition projection (e.g., Schlenker, 2008). The view presented in this paper suggests that this is not accidental: perhaps the formation of precedence relations is not solely driven by requirements at PF, but functions to order nodes for grammatical operations more generally.

AI-use declaration

The author used an AI assistant for proofreading. All analyses, claims, examples, judgments, and final wording are the author's own work.

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